Workflow:
RPI GUI → SKR 1.3 Custom Firmware → XYZ Stepper Motors + Solenoid Valves → Pressure/vacuum dispensing head

Pressure/vacuum source → fitting → Solenoid → fitting → tubing → Barb to luer lock → luer lock tip

Parts to purchase:
Solenoid:
Qolekog DC 12V Solenoid Valve 1/4" NPT 2 Way Normally Closed Air Valve Electric Solenoid Valve

- 0-101 Psi
- 12V operating voltage

Fittings:
Toolly Pneumatic Male Straight 1/4 Inch Tube OD x 1/4 Inch NPT Thread Push to Connect Fittings PC-1/4-N2 (Pack of 10)

- Connect air supply to solenoid input

Tubing: 1/4"
SNS Pneumatic Od 1/4" 10 Meters Orange Color PU Pneumatic Air Tubing Pipe Hose for Air line or Fluid Transfer

- 3.18mm ID

Barb to luer lock:
MEETOOT 10pcs Luer-to-Barb Adapter Male Luer with Lock Ring 1/8" Hose Barb Screw Quick Connector Plug 3.2mm for Laboratory Biochemical/Analytical Instruments and Food Machinery

1/4" NPT to luer lock
MALE LUER TO 1/4" NPT MALE, PLATED BRASS → takes a week to arrive
Could 3d print this piece

- Maybe don't buy it, get adapter above instead

Tips: Blunt straight
18G: PATIKIL Blunt Tip Dispensing Needles 1 Inch, 100 Pcs 18G Industrial Liquid Dispenser Needle with Lure Lock for Refilling Liquid Inks Adhesive Glue
27G: uxcell Blunt Tip Dispensing Needles, 150 Pcs 0.5 Inch 27G Glue Dispensing Stainless Steel Needle Tip Liquid Dispenser for Industrial Oil or Glue Applicator, Light Gray
30G: uxcell Blunt Tip Dispensing Needles, 200 Pcs 0.5 Inch 30G Glue Dispensing Stainless Steel Needle Tip Liquid Dispenser for Industrial Oil or Glue Applicator, Light Yellow

Tips: Blunt angle
25G: PATIKIL Blunt Tip Dispensing Needles with Bent Blunt Needle, 100 Pcs 0.4 Inch 25G Industrial Liquid Dispenser Needle for Refilling Liquid Ink Adhesive Glue, Black
30G: PATIKIL Blunt Tip Dispensing Needles with Bent Blunt Needle, 100 Pcs 0.5 Inch 30G Industrial Liquid Dispenser Needle for Refilling Liquid Ink Adhesive Glue, Pin

Other: (do NOT purchase)

Set:

Air Hose Pipe Tube Kit 6mm OD with 1/4” Push to Connect Air Fittings (13 PCS), Pneumatic Air Line Connector

1N4007 Diode, 1000V 1A High Voltage Rectifier Diode Rectifying Diodes Electronic Diodes, Diodes Electronic Kit (Pack of 100pcs) : Amazon.ca

- Flyback diode

T connector:

TRILINK 1/4" OD Push to Connect Fittings – Air Line Tee Connectors, Quick Connect Pneumatic Fittings, Push Lock Air Hose Fittings for Compressor, Air Tools & Tubing (10 PCS)

- Connect 2 solenoid outputs to attach to dispensing tip